

Homework #2

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Question #1

- Valid Operation
- Valid Operation
- Key Constraint (Key #4 is already taken)
- Key Constraint, Not Null Constraint (Key cannot be null)
- Valid Operation
- Valid Operation
- Integrity Constraint (Did not update WORKS_ON or DEPENDANT)
- Integrity Constraint (Did not update WORK_ON)
- Valid Operation
- Integrity Constraint (Did not update WORKS_ON)
- Valid Operation

Question #2

Grader note: the rho (ρ) is too small in L^AT_EX to be distinguished from a subscript, so instead P is used in it's place.

- $\Pi_{\text{Fname, Minit, Lname}}(\sigma_{\text{Dno} = '5' \wedge \text{hours} = > 10}(\text{roject} \end{verb} \bowtie _{{\text{Dname} = `rojectX' or$
- $\Pi_{\text{Fname} = \text{Dependent_Name}}(\text{mployee} \end{verb} \bowtie \begin{verb} \text{Dependent} \end{verb}))\$$
- $\Pi_{\text{Fname, LName}}(\text{mployee} \end{verb} \div \Pi_{\text{Super_SSN}})(\sigma_{\text{Fname} = `Frankli$
- $\Pi_{\text{Lname, Sum_Hours}}(s_{\text{name}} \mathcal{F}_{\text{sum hours}}(\text{ROJECT} \end{verb} \bowtie \rho_{\text{ESSN} = \text{SSN}}) \begin{verb}$
- $\text{mployee1} \end{verb} \rightarrow \begin{verb} \text{mployee} \bowtie \rho_{\text{ESSN} = \text{SSN} or \text{ork_On} \end{verb} \$ \Pi_{\text{Lname,}}$

- f. $\Pi_{Lname}(\Pi_{Lname, SSN} \bowtie \Pi_{\text{works_On}}) - \Pi_{\text{works_On}}(\Pi_{Lname, SSN})$
- g. $\Pi_{Fname, Average Salary}(\Pi_{Dname, P_dno} \bowtie \Pi_{Department})$
- h. $\Pi_{Sex}(\Pi_{Average Salary}(\Pi_{Sex} = 'female'))$
- i. $\Pi_{Department1} \rightarrow \Pi_{\text{Number} = Dno} \bowtie \Pi_{\text{Number} = Dno}$
- j. $\Pi_{Lname}(\Pi_{SSN} \bowtie \Pi_{MGR_SSN}) - \Pi_{Lname}(\Pi_{SSN})$

Question #3

- $\Pi_{\text{order_id}}(\Pi_{\text{order_id}} \bowtie \Pi_{\text{order_id}})$
- $\Pi_{\text{order_id}}(\Pi_{\text{order_id}} \bowtie \Pi_{\text{order_id}})$
- $\Pi_{\text{order_id}}(\Pi_{\text{order_id}} \bowtie \Pi_{\text{order_id}})$
- $\Pi_{\text{order_id}}(\Pi_{\text{order_id}} \bowtie \Pi_{\text{order_id}})$

Assumptions

- Table headers with the same name are not unique.